

Assignment # 2

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Subject: Human Computer Interaction

Chalo Rider App Redesign

1. Selected Application

This report redesigns **Chalo Rider**, the rider app of Chalo, a ride-booking service used in Mirpur, AJK, for rickshaws, cars and vans. Riders set a pickup and drop-off, choose a vehicle type, wait for a driver and pay in cash.

Chalo Rider was selected for three reasons:

- **It is widely used by the target audience.** Students, staff and residents around MUST use it for daily trips, so real users were easy to reach.
- **Its current UI has clear room for improvement.** A walkthrough of the app showed problems in ride selection, driver identification, address display and error handling (Section 2).
- **Its core flow is short and testable.** Booking a ride is one end-to-end task, which suits a high-fidelity prototype and usability testing.

2. User Research

Research found eight usability problems in the current Chalo Rider app, led by confusing ride selection, unidentifiable drivers and dead-end errors. It combined a walkthrough of the live app with short interviews of riders.

2.1 Target audience

- **Students and staff around MUST** making short daily trips between home, hostels and campus
- **Working professionals in Mirpur** commuting at rush hour
- **Older or occasional riders** with low confidence using apps

Most use Android phones, pay in cash and are price sensitive.

2.2 Research method:

Method	Participants	Purpose
Heuristic walkthrough of the current app	Researcher; 10 screens captured from a real booking	Find usability problems against Nielsen's 10 heuristics
Short interview	User of Chalo Rider App	Find the actual issues the user faces

2.3 Current app (before redesign)

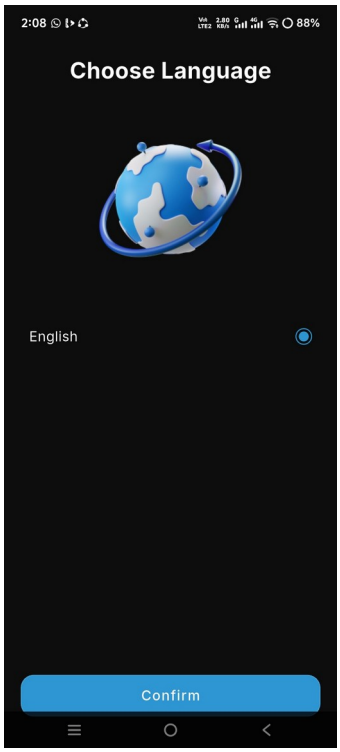


Figure 1: Current app: Choose language

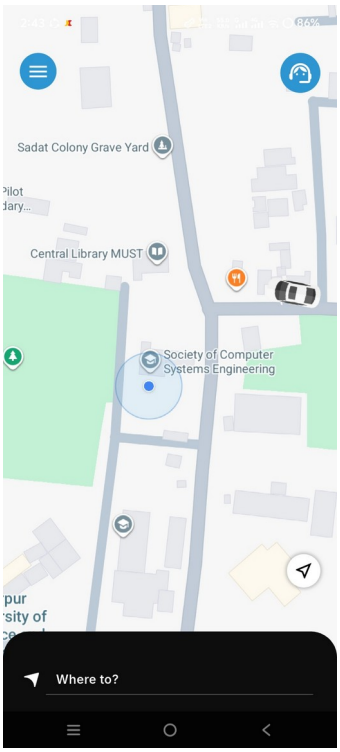


Figure 2: Current app: Home

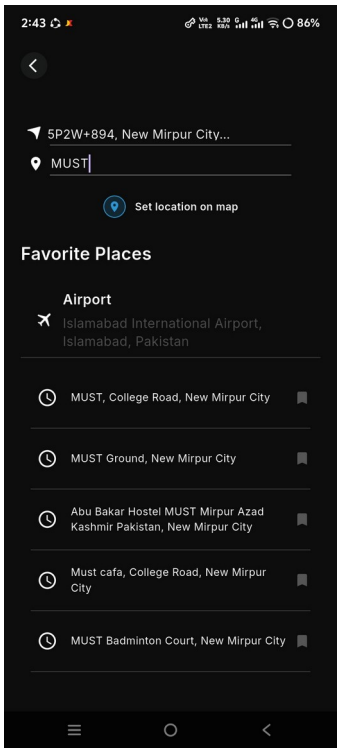


Figure 3: Current app: Destination search



Figure 4: Current app: Confirm pickup and drop-off

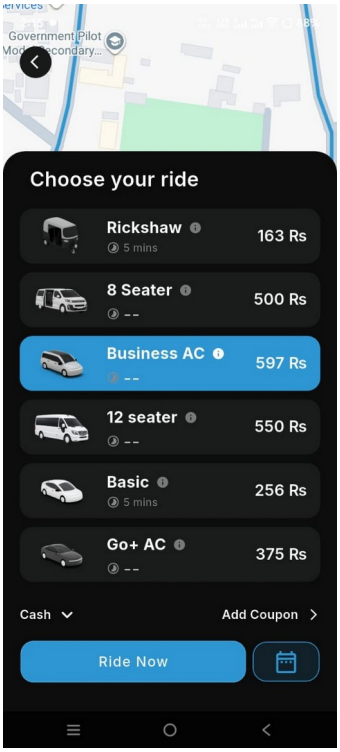


Figure 5: Current app: Choose your ride

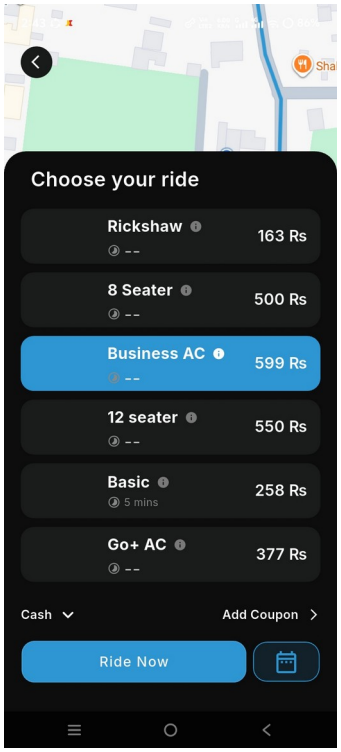


Figure 6: Current app: Choose your ride (vehicle images missing)

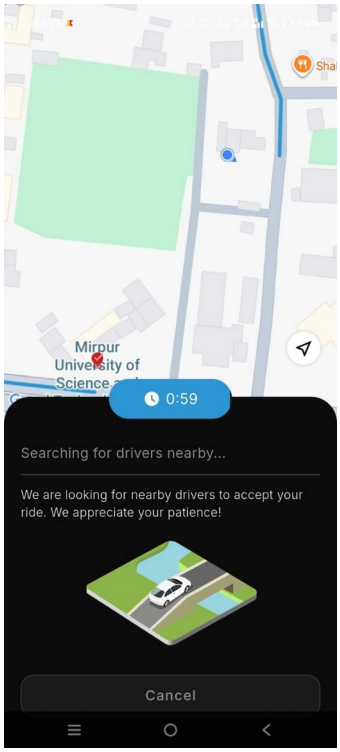


Figure 7: Current app: Searching for drivers

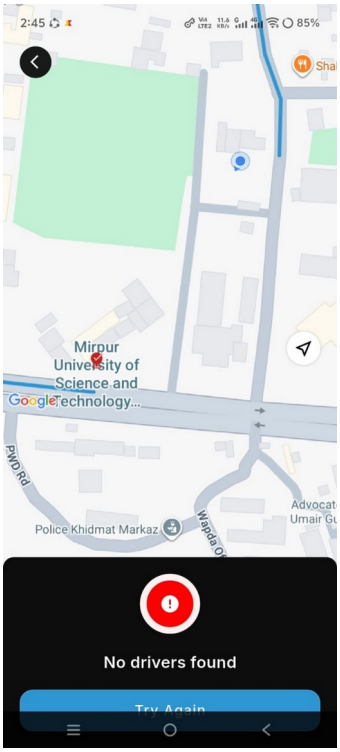


Figure 8: Current app: No drivers found

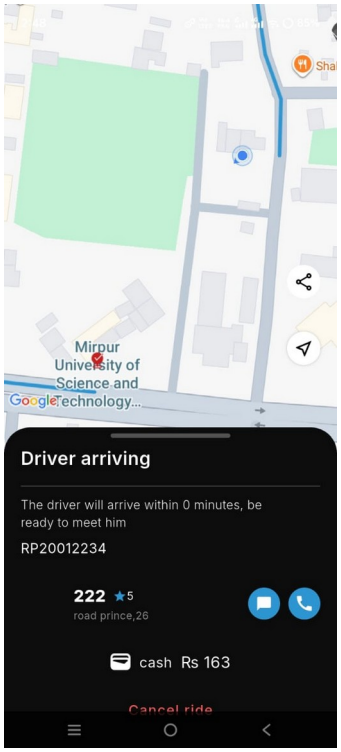


Figure 9: Current app: Driver arriving

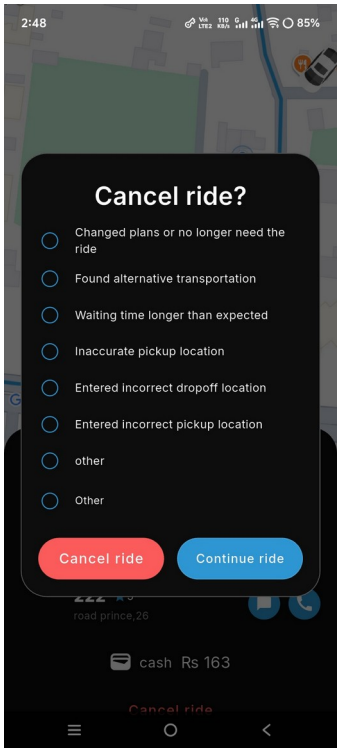


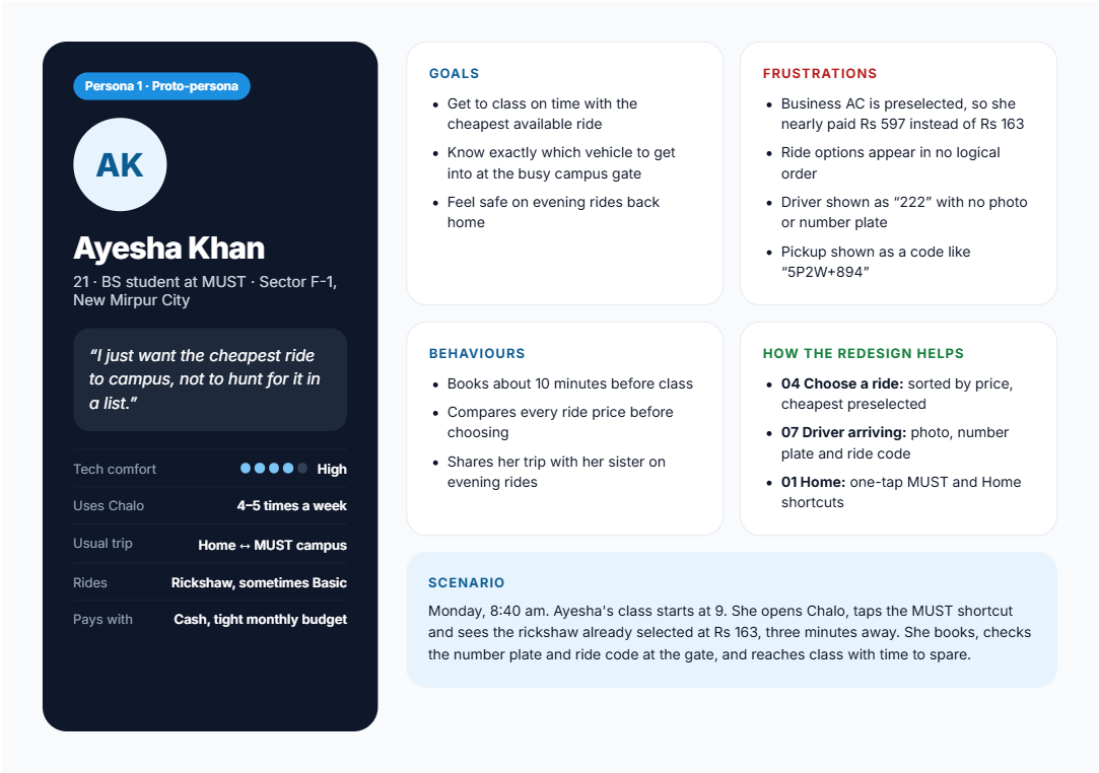
Figure 10: Current app: Cancel ride

2.4 Findings: pain points

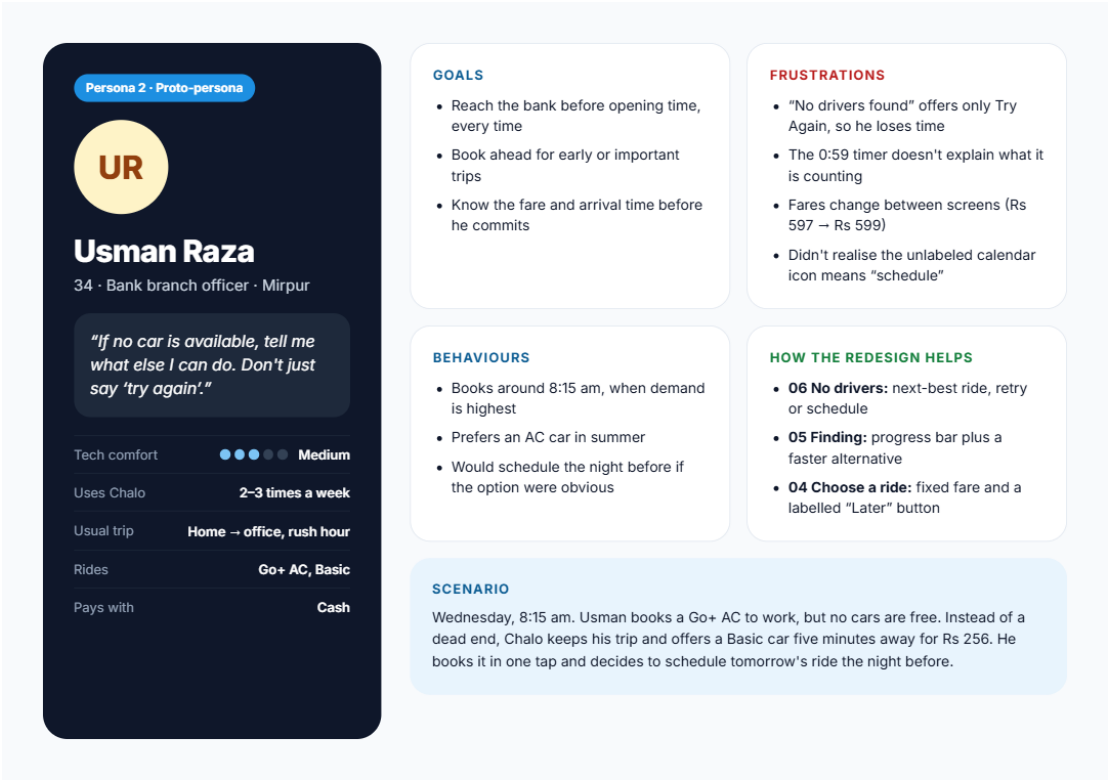
#	Pain point	Evidence in the current app	HCI principle affected
1	Confusing ride selection	Six options in random price order (163, 500, 597, 550, 256, 375 Rs); the most expensive (Business AC) is preselected; most ETAs show “–”	Error prevention
2	Driver hard to identify	Driver shown as “222” with no photo or number plate; “will arrive within 0 minutes”	Error prevention (safety)
3	Unreadable addresses	Pickup shown as plus code “5P2W+894”; recent searches listed under “Favorite Places”	Consistency
4	Dead-end error	“No drivers found” offers only Try Again, partly hidden by the nav bar	Help users recover from errors
5	Unclear waiting state	“0:59” timer with no explanation; no ride summary while searching	Visibility of system status
6	Hidden and unlabeled controls	Main features in a hamburger menu; unlabeled calendar, “+” and share icons	Recognition over recall; affordance
7	Poor accessibility	Low-contrast grey text; pickup and drop-off told apart by colour only; buttons under the nav bar; English is the only language	Accessibility
8	Cluttered cancel dialog	Eight reasons, including both “other” and “Other”	Consistency

3. Personas

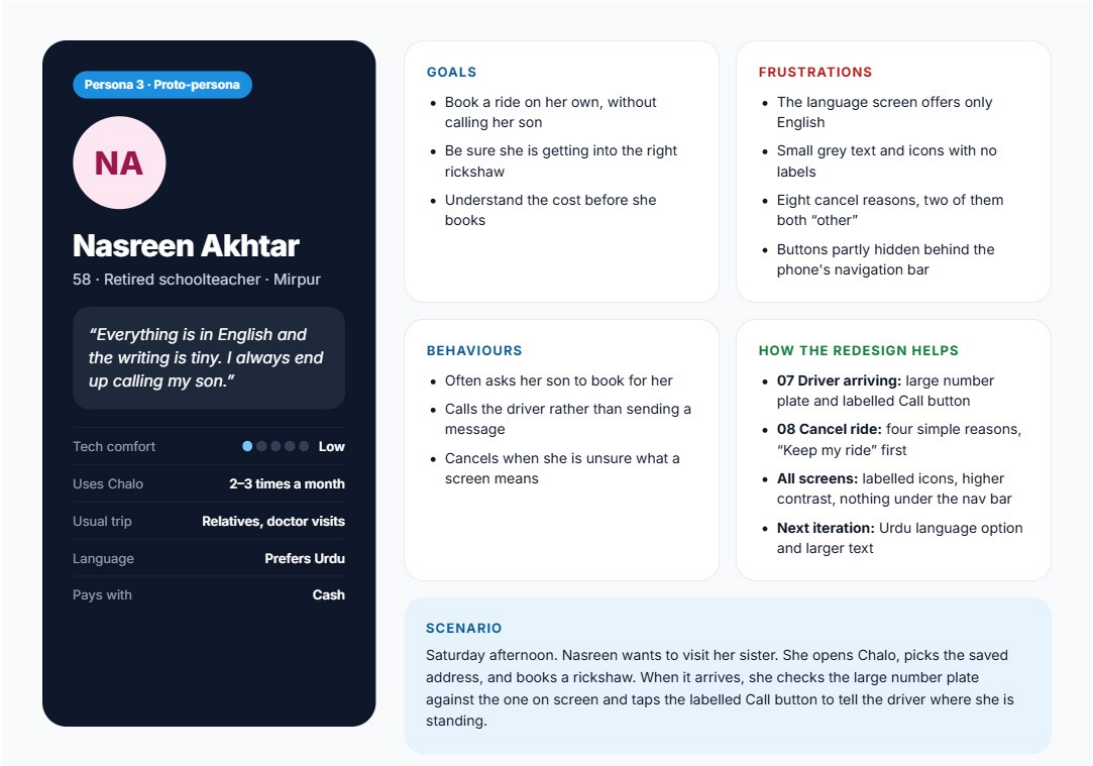
Ayesha Khan represents the largest rider group: budget-conscious MUST students making short daily trips. Every design decision in Section 7 is traced back to her needs.



Usman Raza represents working professionals who commute at rush hour and need a ride that turns up on time. The "No drivers" recovery screen, the search progress bar and ride scheduling in Section 7 are traced back to his needs.



Nasreen Akhtar represents older riders with low tech comfort who prefer Urdu. The labelled icons, simpler cancel options and contrast fixes in Section 7 are traced back to her needs.



4. Creative Ideation: UI Design Concepts

Three layout concepts were explored; the final design combines the map-first base of Concept A with the shortcuts of Concept B and the sorted comparison of Concept C.

4.1 Concepts explored

Concept	Core idea	Strength	Weakness	Decision
A. Map + bottom sheet	Map always visible; each step slides up in a white sheet	Pickup, route and driver stay visible; familiar pattern	Map takes space from content	**Chosen as base**
B. Shortcut-first	Home opens on saved places and last trip, booking in two taps	Fastest for repeat trips like Ayesha's	No map to confirm pickup accuracy	**Shortcuts adopted** into Home
C. Price-first comparison	All vehicles shown as a comparison list with price, seats and AC	Easy to compare fares	Long list for low-confidence users	**Sorted list adopted**; unavailable rides grouped away

4.2 Innovative features in the final design

1. **Smart default:** the cheapest *available* ride is preselected, never the most expensive.
2. **Grouped unavailable rides:** vehicles with no nearby driver collapse into one line instead of looking selectable.
3. **Ride code verification:** a 4-digit code the rider says before boarding, together with the driver's photo and a large number plate.
4. **Smart alternative while waiting:** if a faster vehicle type is nearby, the searching screen offers a one-tap switch.
5. **No dead ends:** when no driver is found, the trip is kept and the next-best ride, a retry and scheduling are offered.
6. **Live availability on Home:** "6 rickshaws and 3 cars near you" before the rider even starts.

7. **Fixed-fare badge:** the price shown is the price paid, removing surprises such as Rs 597 changing to Rs 599.

5. High-Fidelity Prototype

The prototype was built in Figma as 8 clickable screens covering the full booking flow, including the error and cancellation paths.

Prototype link: https://www.figma.com/proto/kdbBNVHtjMNYoRkersrr9P/ChaloRider-Redesign-%E2%80%93-HCI-Assignment-2?node-id=17-5&p=f&t=G3Nv7Ecf3pftYPNb-1&scaling=min-zoom&content-scaling=fixed&page-id=12%3A2&starting-point-node-id=12%3A47

5.1 Screens

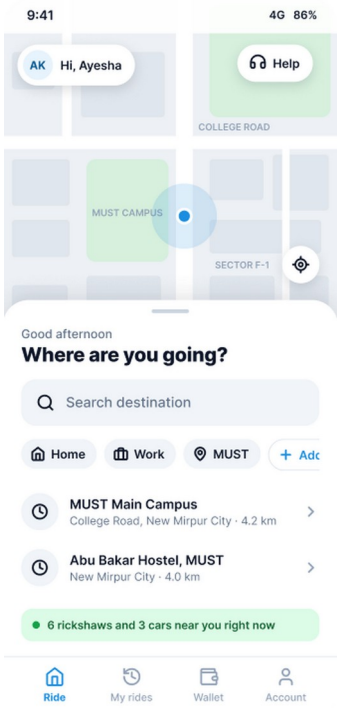


Figure 11: Redesigned: 01 Home

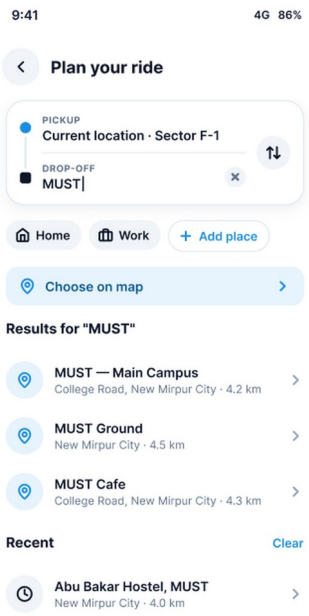


Figure 12: Redesigned: 02 Plan your ride

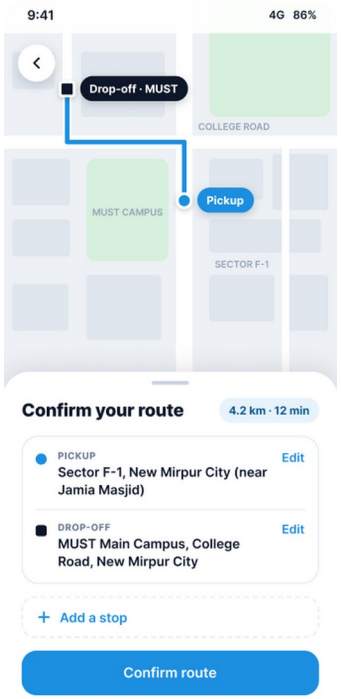


Figure 13: Redesigned: 03 Confirm route

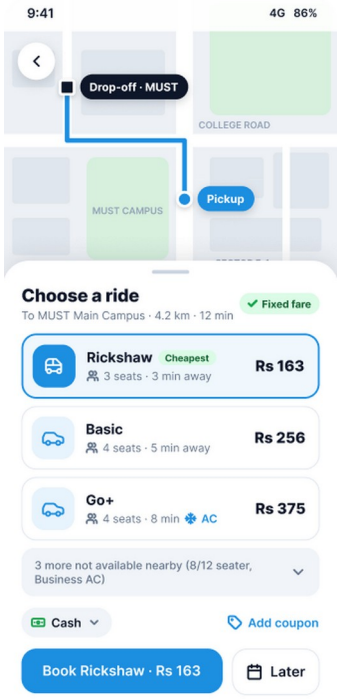


Figure 14: Redesigned: 04 Choose a ride

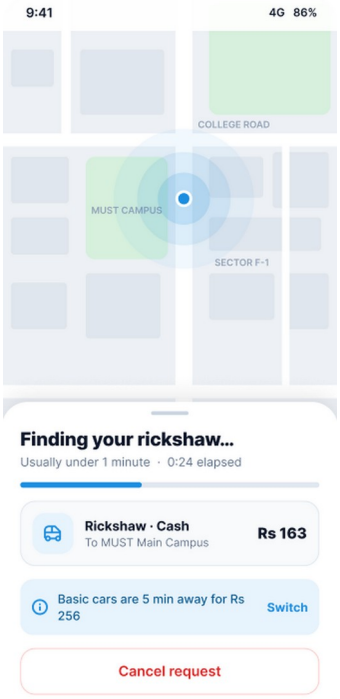


Figure 15: Redesigned: 05 Finding a driver

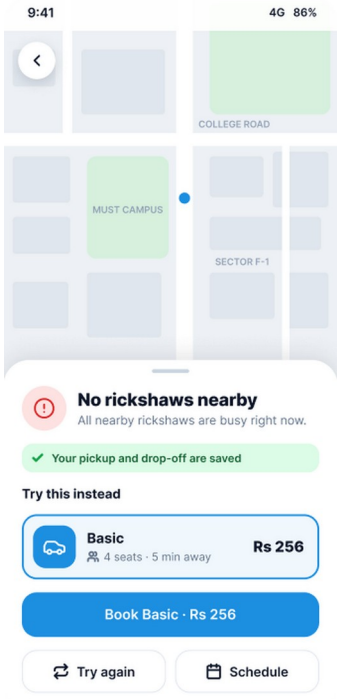


Figure 16: Redesigned: 06 No drivers nearby

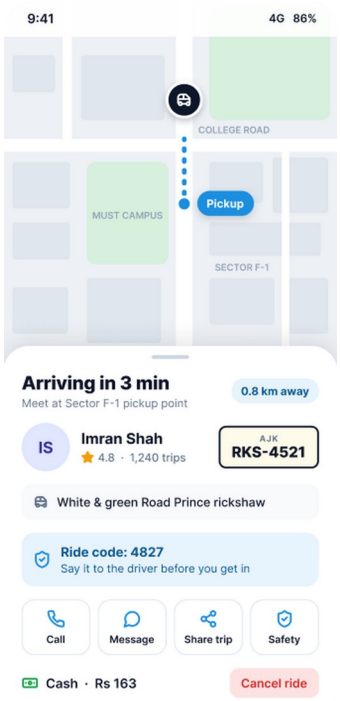


Figure 17: Redesigned: 07 Driver arriving

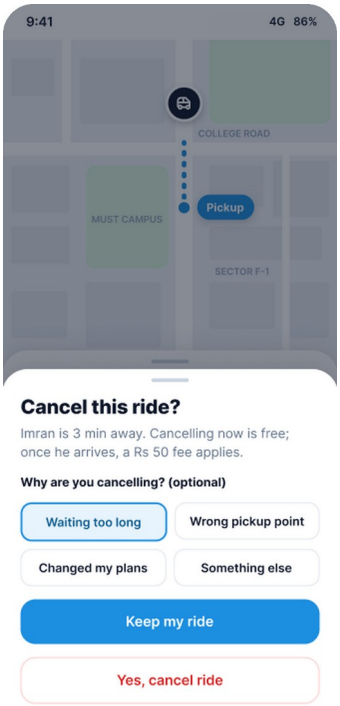


Figure 18: Redesigned: 08 Cancel ride

5.2 Key interactions

From screen	User action	Goes to
01 Home	Tap “Search destination”	02 Plan your ride
01 Home	Tap a recent place (MUST Main Campus)	03 Confirm route
02 Plan your ride	Tap a search result	03 Confirm route
03 Confirm route	Tap “Confirm route”	04 Choose a ride
04 Choose a ride	Tap “Book Rickshaw · Rs 163”	05 Finding a driver
05 Finding a driver	Wait 3 seconds (automatic)	06 No drivers nearby
05 Finding a driver	Tap “Switch” on the faster-car tip	07 Driver arriving
06 No drivers nearby	Tap “Book Basic · Rs 256”	07 Driver arriving
06 No drivers nearby	Tap “Try again”	05 Finding a driver
07 Driver arriving	Tap “Cancel ride”	08 Cancel ride
08 Cancel ride	Tap “Keep my ride”	07 Driver arriving
08 Cancel ride	Tap “Yes, cancel ride”	01 Home
02, 03, 04, 06	Tap Back	Previous screen

The prototype starts on 01 Home (flow “Book a ride”) and uses Smart Animate transitions, so the map stays in place while the bottom sheet changes.

6. Usability Testing and Iteration

In a hypothetical usability-testing scenario, participants tested the Figma prototype on a phone. The results below are hypothetical examples, not measurements from completed sessions.

6.1 Method

- **Participants:** 3 Chalo riders (2 MUST students, 1 staff member)
- **Setup:** Figma prototype on phone, one session per participant

- **Measures:** task completion, time on task, number of errors, and a short comment after each task

6.2 Tasks and results

Task	Completed	Avg. time	Errors	Observations (hypothetical)
1. Book the cheapest ride from home to MUST	3/3	45 sec	1	All selected the cheapest ride; one took extra time to compare fares
2. Identify the correct driver and vehicle	3/3	22 sec	1	All identified the driver and vehicle; one hesitated while checking the details
3. No drivers are available; get a ride anyway	2/3	38 sec	2	found the alternative ride; was unsure how to proceed
4. Cancel the ride, then change your mind	2/3	32 sec	2	completed the flow; struggled to find how to keep the ride

6.3 Feedback and design changes

Participant feedback (hypothetical)	Change made to the design	Screen
“I want to compare the prices easily before selecting a ride.”	Show each fare in larger bold text and keep the “Cheapest” tag on the preselected ride	04 Choose a ride
“I want to confirm that I am getting into the correct vehicle.”	Enlarge the number plate further and show vehicle colour and model on their own line	07 Driver arriving
“What should I do if no drivers are available?”	Place the alternative ride at the top of the sheet as the only primary button, with “Your pickup and drop-off are saved” directly above it	06 No drivers nearby

6.4 Updated screens after usability testing

After the usability tests, the prototype was updated in response to the hypothetical participant feedback in Table 6.3. Figures 19–21 show the updated screens.

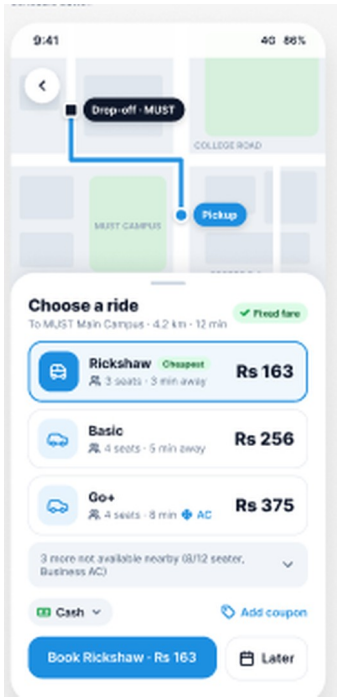


Figure 19: Updated after testing: 04 Choose a ride (larger fares)

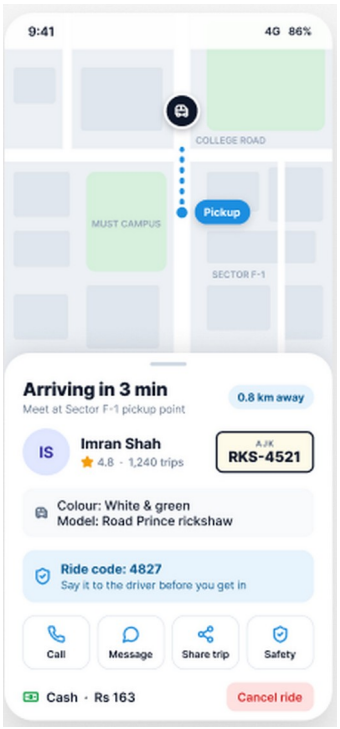


Figure 20: Updated after testing: 07 Driver arriving (larger number plate; colour and model on separate lines)

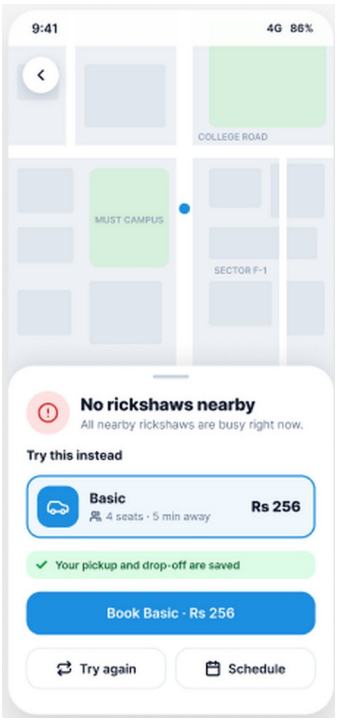


Figure 21: Updated after testing: 06 No drivers nearby (alternative ride placed above the saved-trip message and the booking button)

7. Design Rationale and HCI Justification

Each design decision answers a pain point from Section 2.4 and is justified by a named HCI principle.

7.1 Screen-by-screen rationale

Screen	Design decision	Fixes pain point	HCI principle	Persona served
01 Home	Home, Work and MUST shortcuts plus recent trips on the first screen	3, 6	Recognition over recall; efficiency	Ayesha: Books her daily trip in two taps
01 Home	Bottom navigation and a labelled Help button replace the hamburger menu	6	Recognition over recall; affordance	Nasreen: Finds rides and wallet without guessing
02 Plan your ride	Place names instead of plus codes; separate “Results” and “Recent” lists	3	Match with the real world; consistency	Ayesha: Recognises “MUST Main Campus” immediately
02 Plan your ride	Pickup and drop-off marked by shape (circle, square) and text labels	7	Accessibility (not colour alone)	Nasreen: Cannot confuse the two points
03 Confirm route	Full addresses wrap instead of truncating; distance and time shown	3, 5	Error prevention; visibility of system status	Ayesha: Checks the trip before paying
04 Choose a ride	Options sorted by price; cheapest available preselected	1	Error prevention	Ayesha: Never overpays by default
04 Choose a ride	Unavailable rides grouped into one line; seats, AC and wait time visible	1, 6	recognition over recall	Nasreen: Fewer, clearer choices
04 Choose a ride	Labelled “Later” button and “Fixed fare” badge	6	Affordance; visibility of system status	Usman: Knows he can schedule; trusts the price

Screen	Design decision	Fixes pain point	HCI principle	Persona served
05 Finding a driver	Progress bar with elapsed time; ride summary stays visible	5	Visibility of system status	Usman: Knows what is happening while waiting
05 Finding a driver	One-tap switch to a faster vehicle type	4	Flexibility and efficiency	Usman: Still reaches work on time
06 No drivers nearby	Explains why, keeps the trip, offers next-best ride, retry and schedule	4	Help users recover from errors	Usman: No dead end, no re-entering details
07 Driver arriving	Driver photo, name, rating, large number plate, vehicle colour, ride code	2	Error prevention; match with the real world	Ayesha: Boards the right rickshaw safely
07 Driver arriving	Labelled Call, Message, Share trip and Safety buttons	6	Affordance; recognition over recall	Ayesha: Shares her evening trip with her sister
08 Cancel ride	Four distinct reasons; fee shown before deciding; “Keep my ride” as primary	8	Consistency; user control and freedom; error prevention	Nasreen: Avoids cancelling by mistake

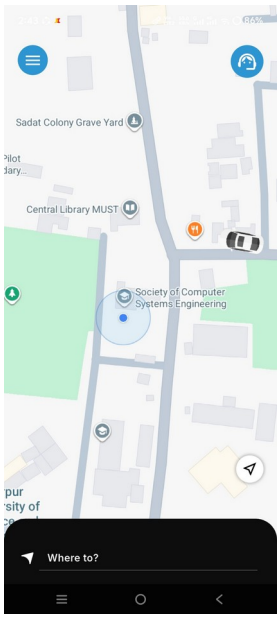
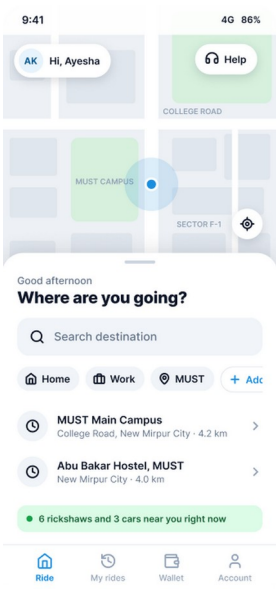
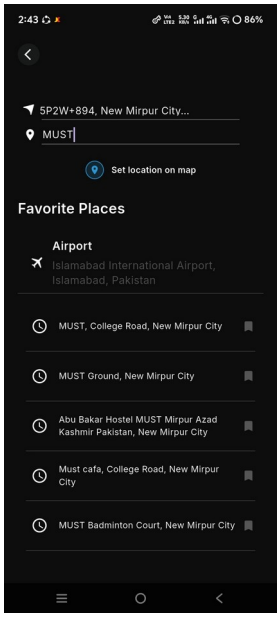
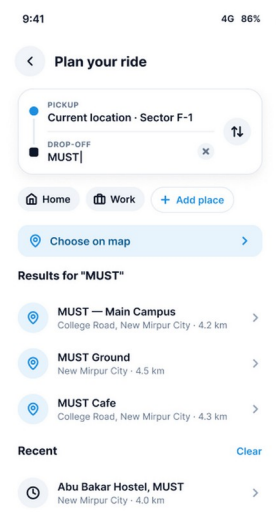
7.2 Principles applied across all screens

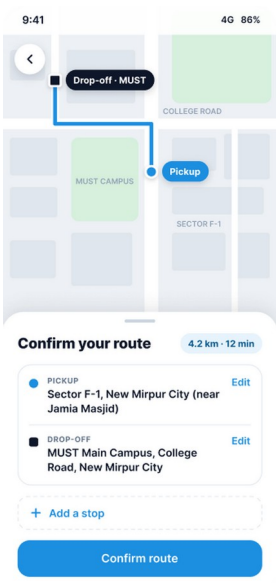
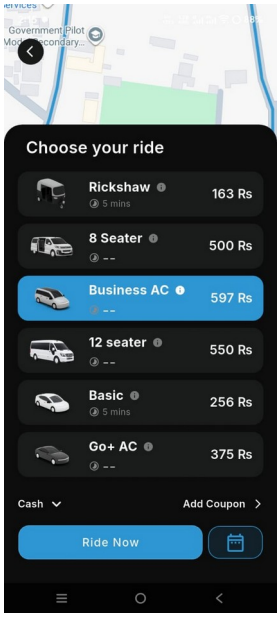
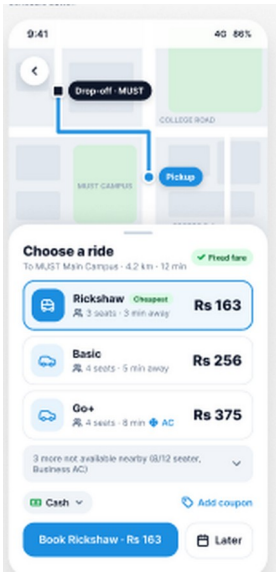
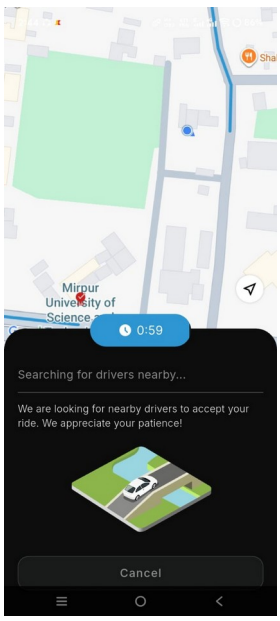
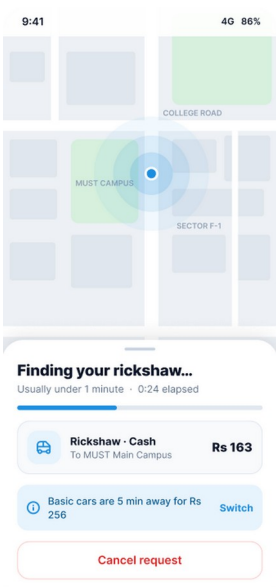
- **Accessibility:** body text uses dark slate (#0F172A) on white, well above the WCAG 4.5:1 contrast ratio; secondary text uses #64748B, which also passes. Touch targets are at least 44 px high, and no control sits under the Android navigation bar.
- **Affordance:** every interactive element looks tappable: filled blue for the main action, outlined for secondary actions, red text only for destructive actions. Every icon has a text label.
- **Consistency:** one visual language on every screen: the map stays in place while a white bottom sheet changes, one blue accent for actions, prices always written as “Rs 163”.
- **Aesthetic and minimalist design:** each screen shows one decision at a time, with only the information needed for that step.

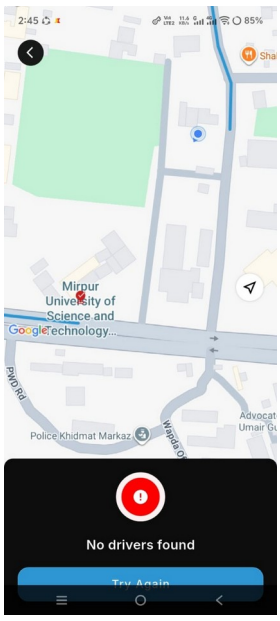
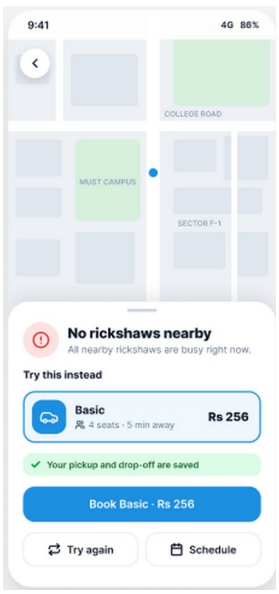
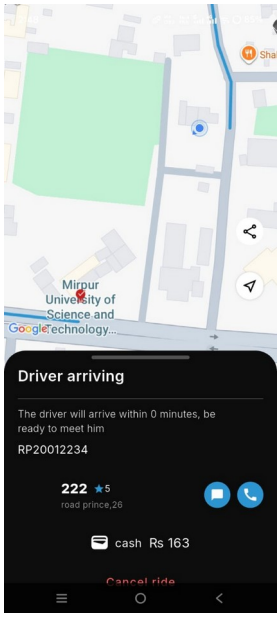
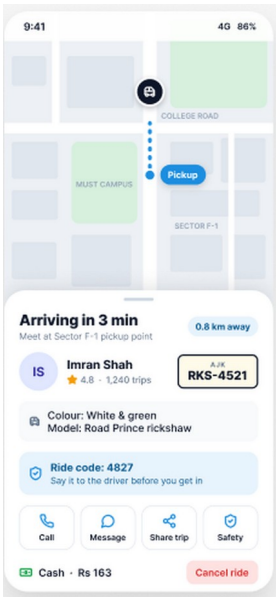
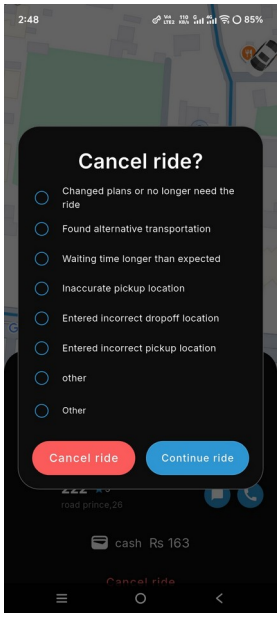
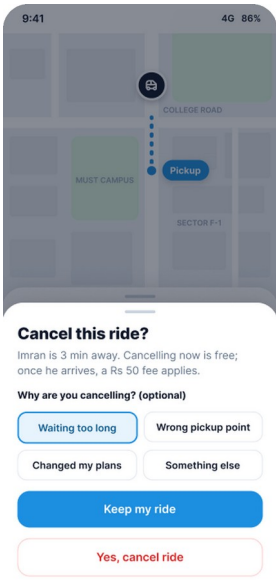
Pain point 7 also noted that English is the only language in the current app. An Urdu language option, needed by older riders such as Nasreen, is planned for the next iteration.

7.3 Before and after comparison

Each current screen is shown next to its redesigned version, with the main changes and the pain points (Section 2.4) they fix. Screens 04, 06 and 07 show the final versions after usability testing.

Before (current app)	After (redesign)	What changed
		**01 Home** • Home, Work and MUST shortcuts plus recent trips replace an empty “Where to?” bar (pain points 3, 6) • Bottom navigation and a labelled Help button replace the hamburger menu (6) • Readable status bar and live availability shown (7)
		**02 Plan your ride** • Place names replace the plus code “5P2W+894” (3) • Separate “Results” and “Recent” lists instead of a mislabelled “Favorite Places” (3) • Pickup and drop-off told apart by shape and label, not colour alone (7)

Before (current app)	After (redesign)	What changed
		**03 Confirm route** • Full addresses wrap instead of being cut off (3) • Distance and time shown before confirming (5) • Labelled “Edit” and “Add a stop” replace unlabelled icons; button sits above the nav bar (6, 7)
		**04 Choose a ride** • Options sorted by price, cheapest available preselected instead of Business AC (1) • Unavailable rides grouped into one line instead of showing “--” (1) • Larger fares, visible seats and AC, “Fixed fare” badge and a labelled “Later” button (1, 6)
		**05 Finding a driver** • Progress bar with elapsed time replaces an unexplained “0:59” timer (5) • Ride summary stays visible while waiting (5) • One-tap switch to a faster ride; Cancel clearly tappable (4, 6)

Before (current app)	After (redesign)	What changed
		**06 No drivers nearby** • Explains why no driver was found instead of a bare error (4) • Next-best ride offered first, with the saved-trip message above the booking button (4) • Try again and Schedule offered; nothing hidden under the nav bar (4, 7)
		**07 Driver arriving** • Driver name, initials avatar and rating replace “222” (2) • Large number plate, vehicle colour and model, and a ride code (2) • Clear ETA instead of “within 0 minutes”; labelled Call, Message, Share and Safety (2, 6)
		**08 Cancel ride** • Four distinct reasons replace eight overlapping ones, including “other” and “Other” (8) • Cancellation fee shown before deciding (8) • “Keep my ride” is the primary button, so cancelling by mistake is harder (8)

8. Expected Impact

The redesign should make booking faster and safer for riders, and turn failed or abandoned bookings into completed trips for Chalo.

Area	Expected impact
User experience	Fewer taps for repeat trips; no accidental expensive bookings; riders can confirm the right vehicle; clear status at every step
Accessibility	Readable text, labelled icons and shape-plus-text cues make the app usable for more riders
Bookings	Recovery options on “No drivers found” keep riders in the app instead of losing the trip
Trust and retention	Fixed fares, honest defaults and driver verification build trust, so riders return
Support load	Clearer screens and visible status mean fewer “where is my driver?” and wrong-vehicle complaints